Project Report: Consumer Financial Complaints End to End Pipeline

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Tools: Python (Pandas/NumPy/Matplotlib/Seaborn), MySQL, Power BI

Dataset: Consumer Financial Protection Bureau (CFPB) Complaints

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1. Phase I: Data Engineering & Preprocessing (Python)

Using two specialized notebooks, I transformed raw, inconsistent data into a high quality analytical dataset.

Step 1: Data Cleaning & Integrity

- **Missing Value Strategy:** I identified high null columns (like *Consumer complaint narrative*) and handled them strategically to maintain dataset volume while ensuring accuracy in the *Sub issue* and *State* fields.
- **Inconsistency Resolution:** I cleaned and standardized categorical values to prevent duplicate groupings (e.g., ensuring "Credit card" and "credit card" were merged).
- **Date Normalization:** I converted the Date submitted and Date received columns into standard datetime formats to allow for accurate time series analysis.

[10]:
```
origin_date = pd.to_datetime('1899-12-30')
date_columns = ['Date submitted', 'Date received', 'Company_Response_Date']

for col in date_columns:
    df[col] = origin_date + pd.to_timedelta(df[col], unit='D')
```

[23]:
```
df.head()
```

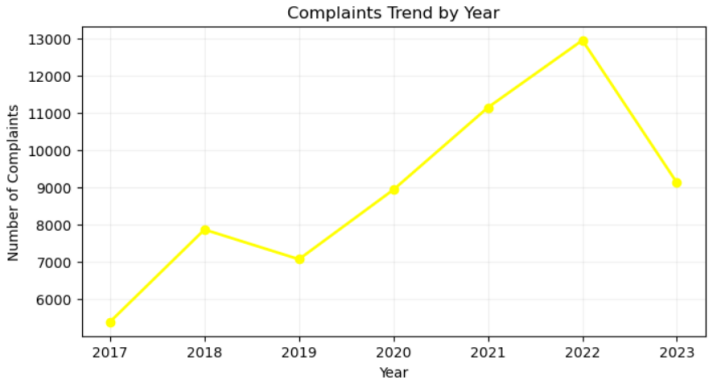
[23]:

	Complaint ID	Submitted via	Date submitted	Date received	State	State_Latitude	State_Longitude	Product	Sub-product	Issue	Sub-issue	Company public response	Company response to consumer	Tim respons
0	4848023	Referral	2021-10-24	2021-10-27	NY	42.165726	-74.948051	Mortgage	Conventional home mortgage	Applying for a mortgage or refinancing an exis...	Not Specified	Company has responded to the consumer and the ...	Closed with explanation	
1	3621464	Web	2020-04-24	2020-04-24	FL	27.766279	-81.686783	Money transfer, virtual currency, or money ser...	Refund anticipation check	Lost or stolen check	Not Specified	Company has responded to the consumer and the ...	Closed with monetary relief	
2	5818349	Web	2022-07-27	2022-07-27	CA	36.116203	-119.681564	Credit reporting, credit repair services, or o...	Credit reporting	Incorrect information on your report	Account information incorrect	Company has responded to the consumer and the ...	Closed with explanation	

Step 2: Feature Engineering & EDA

- **Visual Exploration:** I used **Matplotlib** and **Seaborn** to create heatmaps and distribution plots to identify outliers before moving the data to SQL.
- **Geographic Mapping:** I created a custom mapping function to group 50+ states into **Census Regions** (Northeast, Midwest, South, West). This allowed for higher level geographic insights.
- **Time Features:** I extracted Year and Month features to analyze seasonal trends.

```
[88]: yearly_counts = df.groupby(df['Date submitted'].dt.year).size()
plt.figure(figsize=(8,4))
yearly_counts.plot(kind='line', marker='o', color='yellow', linewidth=2)
plt.title('Complaints Trend by Year')
plt.xlabel('Year')
plt.ylabel('Number of Complaints')
plt.grid(True, alpha=0.2)
plt.show()
```



2. Phase II: Deep Dive Analysis (MySQL)

Once the data was structured, I migrated it to a MySQL environment to perform complex relational queries and performance benchmarking.

Step 1: Advanced Window Functions

- **The "Top 3 Issues" Challenge:** I implemented DENSE_RANK() OVER (PARTITION BY Product ORDER BY COUNT(*) DESC) to solve a complex business problem: identifying the top 3 pain points for *every* unique financial product simultaneously.

```sql
-------------------------------------------------------------------------------
-- Products with their Top 3 Issues
-------------------------------------------------------------------------------
SELECT Product, Issue, Complaint_Count
FROM (
    SELECT
        Product,
        Issue,
        COUNT(*) AS Complaint_Count,
        DENSE_RANK() OVER (PARTITION BY Product ORDER BY COUNT(*) DESC) as Issue_Rank
    FROM financial_complaints
    GROUP BY Product, Issue
) AS Ranked_Issues
WHERE Issue_Rank <= 3
ORDER BY Product, Issue_Rank;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Product	Issue	Complaint_Count
Checking or savings account	Managing an account	30218
Checking or savings account	Closing an account	5906
Checking or savings account	Opening an account	5450
Credit card or prepaid card	Problem with a purchase shown on your statem...	8830
Credit card or prepaid card	Getting a credit card	3734
Credit card or prepaid card	Other features, terms, or problems	3266
Credit reporting, credit repair services, or other...	Incorrect information on your report	8290
Credit reporting, credit repair services, or other...	Problem with a credit reporting company's inves...	3282
Credit reporting, credit repair services, or other...	Improper use of your report	3034
Debt collection	Attempts to collect debt not owed	2702
Debt collection	Written notification about debt	974
Debt collection	Took or threatened to take negative or legal ac...	760
Money transfer, virtual currency, or money ser...	Fraud or scam	3902

Step 2: Operational Analytics

- **Response Benchmarking:** I built queries to segment companies into "Timely" vs. "Late" responders, specifically targeting the 3.94% of non compliant cases.
- **Drill Down Logic:** I wrote multi level GROUP BY queries to analyze complaints at the **Product > Sub product > Issue** level, which served as the logical backbone for the Power BI hierarchy.
- **Monthly Trend Calculation:** I performed aggregations to identify the "July Spike," confirming a consistent 15% increase in volume during the mid year period.

```
--------------------------------------------------------------------------------
-- Month Wise Complaints Each Year
--------------------------------------------------------------------------------
select
    year(`Date submitted`) as Complaint_Year,
    month(`Date submitted`) as Complaint_Month,
    count(`Complaint ID`) as Total_Complaints
from financial_complaints
group by
    year(`Date submitted`),
    month(`Date submitted`)
order by
    Complaint_Year,
    Complaint_Month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

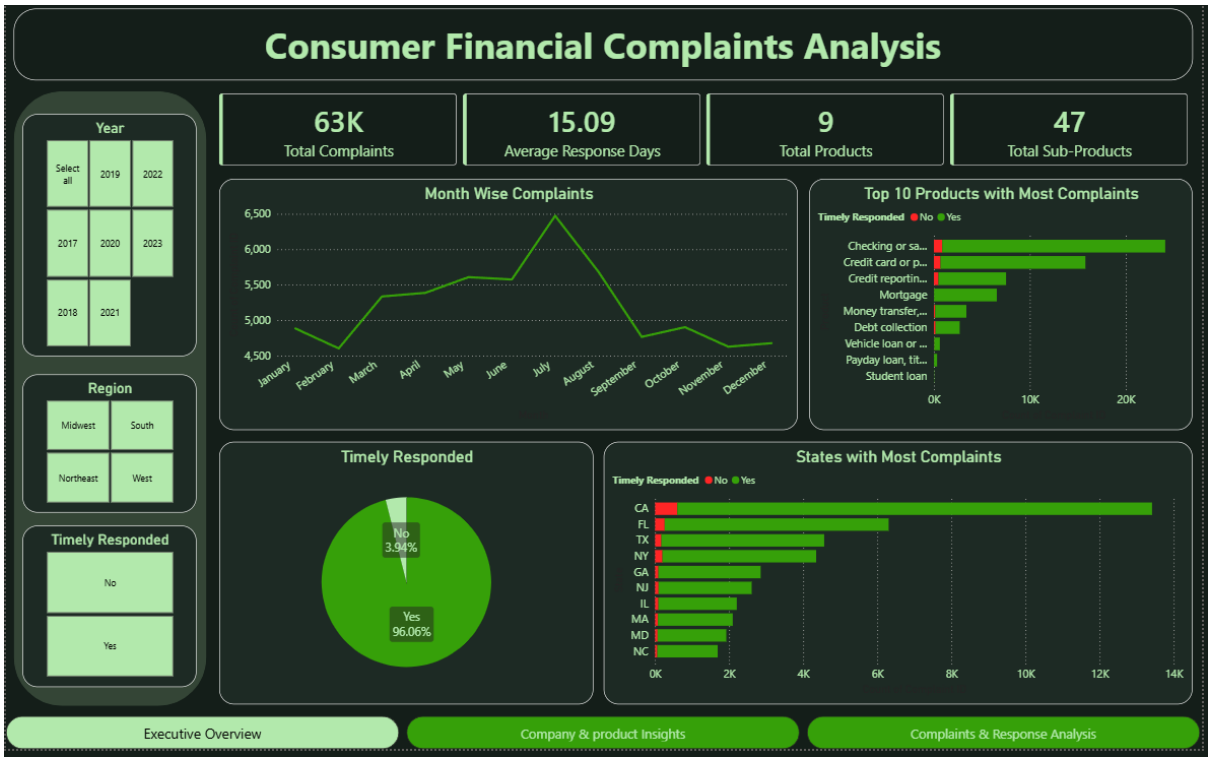
Complaint_Year	Complaint_Month	Total_Complaints
2022	8	2562
2022	9	2388
2022	10	2266
2022	11	2262
2022	12	2290
2023	1	2366
2023	2	2184
2023	3	2458
2023	4	2116
2023	5	2170
2023	6	2036
2023	7	3498
2023	8	1434

3. Phase III: Strategic Visualization (Power BI)

I developed a 3 page, interactive dashboard using a modern "Midnight Green" dark mode theme to communicate findings to stakeholders.

Page 1: Executive Overview

- **Visuals:** KPI cards (63K Complaints, 15.09 Avg Response Days), Line Chart for Monthly Trends, and a State wise Bar Chart.
- **Goal:** Provide an "at a glance" health check of consumer sentiment.



Page 2: Company & Product Insights

- **Visuals:** Horizontal bar charts for Top 10 Sub products and Companies, plus a "Company Tier" Pie Chart (Small, Medium, Large).
- **Goal:** Identify which specific institutions and product lines are driving the most friction.

Page 3: Regional & Response Analysis

- **Visuals:** Regional Donut Chart, Submission Channel Bar Chart, and a Response Type breakdown.
- **Goal:** Analyze *how* complaints are submitted and *how well* they are being resolved geographically.

4. Key Business Findings

1. **Product Friction:** Checking/Savings accounts and Credit Reporting represent the highest risk areas for financial institutions.
2. **Seasonal Trend:** A recurring surge in July suggests a need for increased support staffing during the summer months.
3. **Efficiency Gap:** While the 96% timely response rate is strong, the "Company Tier" analysis shows that smaller firms struggle more with response latency than larger MNCs.

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5. Project Conclusion

The analysis of the Consumer Financial Complaints dataset successfully demonstrates a complete end to end data pipeline. By integrating Python, SQL, and Power BI, I was able to transform a high volume, unstructured dataset into a suite of actionable insights.

The technical workflow proved that **Python** is essential for creating a "clean room" for data, especially when handling complex feature engineering like the Census Region mapping. The subsequent deep dive in **MySQL** enabled the extraction of high level rankings and seasonal trends most notably the July volume surge that would be difficult to pinpoint with basic analysis tools. Finally, the **Power BI** dashboard provides a professional interface for stakeholders to interact with these findings in real time.

Overall, this project highlights the importance of a structured analytical approach. The insights discovered regarding product friction points and geographic complaint density provide a clear roadmap for improving operational efficiency and customer response strategies. This project stands as a verified proof of concept for managing large scale financial data to drive informed business decisions.